

Notes: Equiangular Tight Frames via Geometric Invariant Theory

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2025

Chapter 1

Motivation

Let V be a K vector space where K is algebraically closed.

Definition 1. Let $Fr_n(V) := \{(v_i)_{i=1}^n \in V^n \mid v_i \neq \lambda v_j, \text{ for } i \neq j, \text{ span}(\{v_i\}) = V\}$ be called frame.

We define some actions of different groups on $Fr_n(V) \times Fr_n(V^*)$:

- $GL(V)$ acts on the left as : $M((v_i), (\phi_i)) = ((Mv_i), (v \mapsto \phi_i(M^{-1}v)))$
- $(K^\times)^n \rtimes S_n$ acts on the right as (scaling and permuting frames):
 - $((v_i), (\phi_i))\lambda_i = ((\lambda_i v_i), (v \mapsto \phi_i(\lambda_i^{-1}v)))$
 - $((v_i), (\phi_i))\sigma = ((v_{\sigma(i)}), (\phi_{\sigma(i)}))$

Definition 2. Now we need to look at the subspace of $GL(V) \backslash Fr_n(V) \times Fr_n(V^*) / (K^\times)^n \rtimes S_n$ called **ETB_n(V)** which satisfies the following:

1. $\phi_i(v_i) = 1$
2. $\phi_i(v_j)\phi_j(v_i) = \frac{n-d}{d(n-1)}$ for $i \neq j$
3. $v_j = \frac{d}{n} \sum_{i=1}^n \phi_i(v_j)v_i$

Now the **BIG QUESTIONS** are:

- what is $ETB_n(V)$? Variety?
- when is dimension of $ETB_n(V)$ zero? In that case, what is the number of points $\#ETB_n(V)$?

Special Case:

$K = \mathbb{C}, \phi_i(v) = \langle v_i, v \rangle = \bar{v}_i, v$ is called **ETF** Equiangular Tight Frames...

1. unit norm: $\langle v_i, v_i \rangle = 1$
2. equiangular: $|\langle v_i, v_j \rangle|^2 = \frac{n-d}{d(n-1)}$
3. tight: $v = \frac{d}{n} \sum_{i=1}^n \langle v_i, v \rangle v_i$

Chapter 2

Polarization and Restitution:

2.1 Invariant functions, quotients, and finite generation

2.1.1 Invariant functions.

Let k be an algebraically closed field and let G be an algebraic group acting on an affine variety $X = \operatorname{Spec} k[A]$, where A is a finitely generated k -algebra. The action of G on X is equivalently given by a rational action of G on A by k -algebra automorphisms. We define the ring of *invariant functions* by

$$A^G := \{f \in A \mid g \cdot f = f \text{ for all } g \in G\}.$$

Remark 1 (Geometric meaning). *An invariant function is constant along G -orbits. Thus A^G measures exactly the functions on X that cannot distinguish points inside a single orbit. The philosophy of GIT is that the quotient space should have coordinate ring A^G .*

2.1.2 Spec vs. Specm.

There are two closely related spaces associated to a ring A^G :

$$\operatorname{Spec} A^G \quad \text{and} \quad \operatorname{Spec}^m A^G.$$

The maximal spectrum $\operatorname{Spec}^m A^G$ corresponds to closed points, while $\operatorname{Spec} A^G$ includes also generic points encoding specialization relations.

Remark 2. *In algebraic geometry, $\operatorname{Spec} A^G$ is the correct object for forming quotients. Even when one is interested in closed orbits (points), non-closed orbits play a crucial role, and their closures are naturally encoded in $\operatorname{Spec} A^G$, not in $\operatorname{Spec}^m A^G$.*

Example 1. *Let $G = \mathbb{Z}/2$ act on $A = k[x, y]$ by*

$$\sigma(x) = y, \quad \sigma(y) = x.$$

Then the invariant ring is

$$A^G = k[x + y, xy].$$

The morphism

$$\pi : \mathbb{A}^2 \rightarrow \mathbb{A}^2, \quad (x, y) \mapsto (x + y, xy)$$

is G -invariant and induces an isomorphism

$$\operatorname{Spec} k[x + y, xy] \cong \operatorname{Spec} A^G.$$

Geometrically, π identifies the points (x, y) and (y, x) , and separates distinct orbits except when (x, y) and (y, x) lie in the same orbit closure. In particular, invariant functions separate closed orbits but may fail to separate non-closed ones.

Remark 3 (Key phenomenon). *The fibers of π are precisely orbit closures, not individual orbits. This is the fundamental reason the GIT quotient is a categorical quotient rather than a geometric one.*

2.1.3 Affine schemes and schemes.

An *affine scheme* is a space of the form $\text{Spec} A$ for a ring A . A *scheme* is obtained by gluing affine schemes along open subsets. Thus schemes are locally affine objects.

Remark 4. *For an affine scheme $X = \text{Spec} A$, one has a canonical identification*

$$A \cong \mathcal{O}_X(X).$$

This property characterizes affine schemes among all schemes.

2.1.4 Quotients and finite generation.

Given an affine G -variety $X = \text{Spec} A$, a natural candidate for the quotient is

$$X//G := \text{Spec} A^G.$$

This construction only makes sense if A^G is finitely generated as a k -algebra.

Theorem 3 (Hilbert, classical form). *If G is a reductive algebraic group acting rationally on a finitely generated k -algebra A , then A^G is finitely generated.*

Remark 5 (Why reductivity matters). *Reductive groups have enough invariant linear functionals to control the growth of invariants. For non-reductive groups, invariant rings can be pathologically large.*

2.1.5 Hilbert's 14th problem and Nagata's counterexample.

Hilbert asked whether A^G is finitely generated for *every* linear action of an algebraic group. This is false.

Theorem 4 (Nagata). *There exists a linear action of a unipotent algebraic group G on a polynomial ring A such that the invariant ring A^G is not finitely generated.*

Remark 6 (Conceptual takeaway). *Finite generation is not a formal property of group actions; it is a deep structural fact tied to reductivity. GIT works precisely because it restricts attention to reductive group actions, where quotients exist as affine schemes of finite type.*